From Public Data to a Coupled Data-Center Energy–Water Model

Meta Prineville as an Empirical Test Bed

Advisor Meeting Brief

Prineville Public-Data Baseline v2 — City utility-meter integration — `e0b175e`

Goal for this meeting: understand the big modeling idea, the evidence that currently supports each link, what is reconstructed rather than observed, what has worked or failed, and the next identification step.

1 Big picture: what are we ultimately trying to model?

The long-run research objective is to connect a data center's operation to the physical systems that bear its environmental impacts:

workload → IT power / heat → facility cooling and power → {grid, direct water} → {emissions / generator water, water-system / groundwater state} → operations, siting, and policy.

The literature contains strong models for many individual links, but the links are usually studied separately. Operational papers often treat carbon or water intensity as exogenous signals; siting models often use static regional water coefficients; groundwater models take pumping schedules as given. The opportunity is to connect these layers while keeping the physical and accounting boundaries explicit.

Prineville is the empirical test bed. The present project asks how much of this full chain can be populated, reconstructed, and validated from public data for one real hyperscale campus before moving to coupled optimization.

One view of the full model and current Prineville status

2 Evidence stack: who provides the data, and what boundary do they observe?

The strongest feature of the pipeline is that it does *not* treat all public data as if they measured the same thing. Different authorities observe different physical or accounting boundaries.

Three boundary rules that prevent false precision and double counting

1. **Regional/context quantities are not campus quantities.** For example,
$$E^{Meta}_{campus} \neq D^{PACW} \neq G^{Oregon}_{plants}.$$
2. **Different water boundaries are not summed unless a physical accounting relationship is known.** For example,
$$W^{Meta}_{campus} \neq W^{City}_{production} \neq Q^{direct\,POD} \neq W^{USGS}_{HUC12}.$$

Layer	Main quantity	Current status	What that means in Prineville
Workload	jobs / utilization x_t, u_t	SCENARIO	No public Meta workload telemetry; current workload paths are synthetic.
IT power	P_t^{IT}	FITTED/ SCENARIO	Latent hourly shape; constrained indirectly by annual facility electricity.
Facility electricity	E_y^{fac}	REPORTED	Annual Meta campus electricity is high-confidence ground truth.
Cooling	Q_t^{cool}, P_t^{cool}	DERIVED	Simplified gray-box physics driven by IT load, weather, and assumed parameters.
Weather	T, T_d, T_{wb}, p	MEASURED/ DERIVED	NOAA KS39/KRDM observations; wet bulb derived psychrometrically.
Campus water	W_y^{Meta}	REPORTED	Annual Meta withdrawal is observed from 2014 onward.
Subannual campus water	W_t	Mixed: City-service observed; campus total still reconstructed	City meters observe Facebook Data Center WATER-COMM + ADD'L WATER, not Meta campus-total withdrawal.
Regional grid demand	D_t^{PACW}	MEASURED/ PROXY	EIA-930 when available; FERC-constrained proxy for earlier years.
Grid carbon	CI_t	DERIVED/ PROXY	Regional average/accounting context; not Meta-specific marginal emissions.
Generator water/emissions	$W_{g,t}, CO_{2,g,t}$	REPORTED/ DERIVED	Strong plant context; no claim that particular plants physically serve Meta.
Water-system pumping	$Q_{i,t}$	REPORTED	OWRD municipal/direct-POD pumping at specific reporting boundaries.
Groundwater state	$h_{i,t}$	MEASURED, QC-qualified	GWIS now provides measured levels; measurement-condition uncertainty remains.
Groundwater response	pumping $\rightarrow h$	NOT IDENTIFIED	Next experiment: test whether antecedent pumping improves prediction beyond season/trend.
Optimization	workload/siting/source choice	NOT IDENTIFIED	Future layer; response functions and costs are not yet sufficiently identified.

Table 1: The general model and the current empirical status of each link.

3. **Geographic overlap does not imply physical connection.** A HUC12 is a geographic/surface-hydrology unit; it is not automatically a groundwater node, and proximity alone does not identify a serving well, generator, or aquifer connection.

3 The pipeline in plain language

The implementation is best understood as a sequence of evidence decisions rather than as a collection of scripts:

$\boxed{\text{authoritative raw source}} \rightarrow \boxed{\text{source-specific QC}} \rightarrow \boxed{\text{identity + boundary crosswalk}}$

$\rightarrow \boxed{\text{canonical quantity}} \rightarrow \boxed{\text{model / reconstruction if needed}} \rightarrow \boxed{\text{independent validation}} \rightarrow \boxed{\text{scenario / decision mode}}$

A multi-timescale problem

The model cannot honestly force every quantity onto the same time resolution:

Layer	Natural / available scale
Workload / server telemetry	seconds–minutes, but unavailable for Meta
Facility cooling / weather	minutes–hours
Regional grid	hourly
Meta campus disclosures	annual
OWRD pumping	monthly
Groundwater response	monthly / seasonal for the first reduced model
Siting / planning	annual / multi-year

The scientific challenge is therefore not merely filling missing cells; it is translating between time scales without pretending that annual measurements identify hourly internal states.

Canonical KS39/KRDM weather covers 122,736 unique local-calendar hours for 2011–2024. KS39 is preferred from 2015-09-01 local when QC-usable; KRDM is the backbone and gap-fill. Wet-bulb is derived psychrometrically.

4 From source to quantity to model

The provenance vocabulary used throughout the project is deliberately small:

Status	Meaning
REPORTED	Value officially reported by an operator or agency for the stated boundary.
MEASURED	Direct observational measurement such as weather or groundwater level.
DERIVED	Computed from observed/validated quantities using an accounting or physical relationship.
FITTED	Estimated from data using a documented statistical or physical model.
PROXY	Substitute for an unavailable target; useful structurally but not claimed as the target measurement.
SCENARIO	Explicitly assumed or simulated quantity used for sensitivity/counterfactual analysis.
NOT IDENTIFIED	Scientifically relevant quantity/relationship that current evidence does not determine.
NOT NECESSARY	Not an active acquisition target for the present public-data pipeline; not necessarily scientifically unimportant.

5 The model: general structure and current Prineville approximation

The literature suggests a coherent coupled spine. The Prineville implementation intentionally uses the simplest version supported by available public data.

5.1 Workload → IT power

A general operational model would begin with

$$E_t^{IT} = \mathcal{P}(x_t, u_t, \text{hardware}, \text{workload})\Delta t.$$

Prineville: workload and server telemetry are not public. P_t^{IT} is therefore latent/scenario-based, not measured.

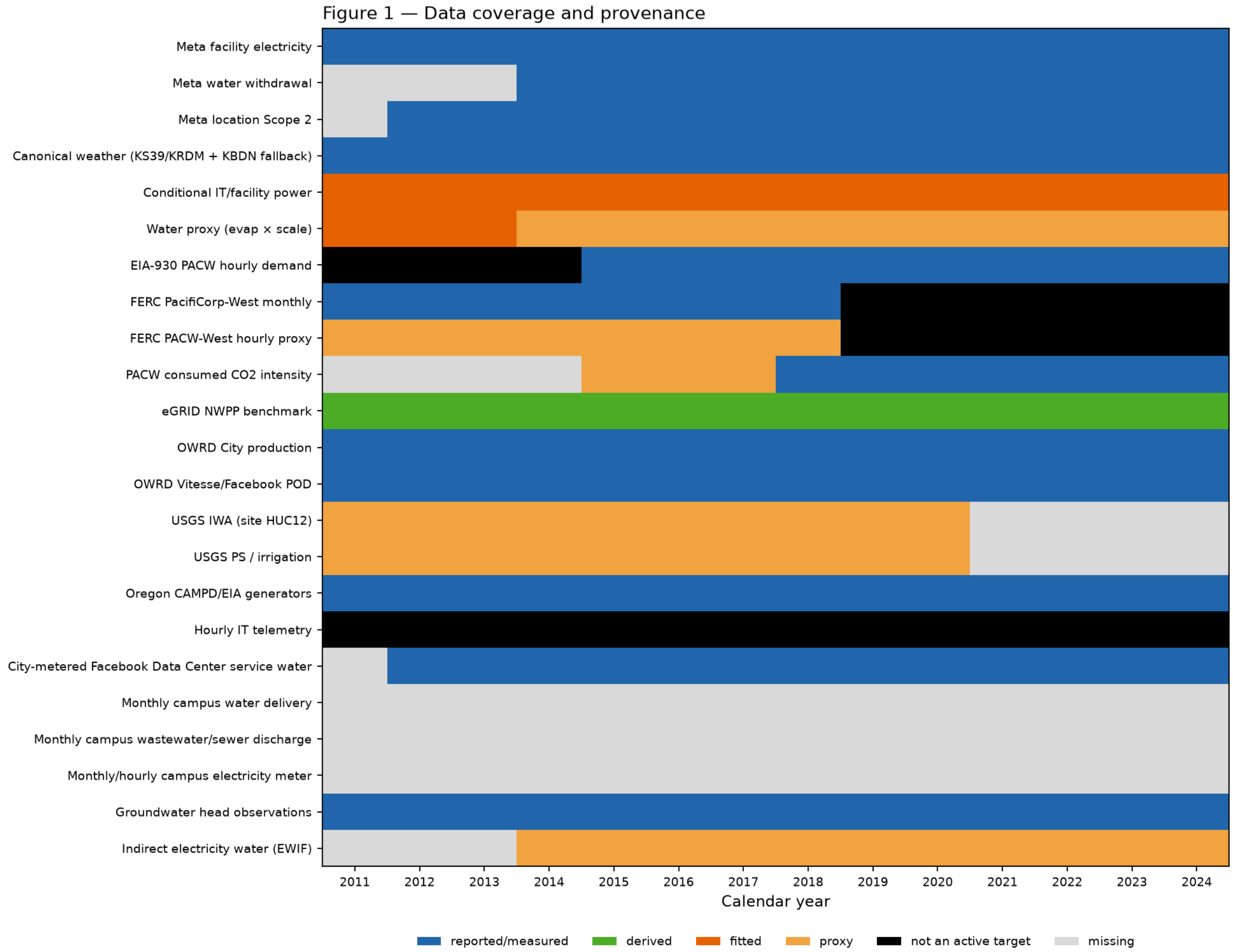

Figure 1: Year-by-year coverage status, not a data-density plot. A MEASURED/REPORTED cell means a source or measurement exists for that calendar year. For groundwater, that means at least one valid GWIS observation in the year—not monthly completeness or network completeness. PROXY is a model-backed substitute, not a measurement. NOT NECESSARY means not an active acquisition target for the current public-data pipeline, not that the quantity is scientifically useless. Missing remains an active gap.

5.2 IT power → facility electricity and cooling

The current facility balance is

$$P_t^{fac} = P_t^{IT} + P_t^{fan} + P_t^{other} + P_t^{cool,aux}, \qquad PUE_t = \frac{P_t^{fac}}{P_t^{IT}}.$$

The annual IT scale is chosen so that

$$\sum_{t \in y} P_t^{fac} \Delta t = E_y^{Meta}.$$

Interpretation: this is an annual-closed reconstruction. The exact annual match is an accounting/calibration closure, *not* electricity forecast accuracy.

5.3 Cooling → direct water

The general relationship is

$$W_t^{dir} = \mathcal{W}\Big(Q_t^{cool}, T_t^{wb}, m_t, u_t^{cool}\Big),$$

where m_t denotes cooling technology/mode.

The current Prineville gray-box estimates a raw evaporative requirement from weather, airflow, heat load, and simplifying cooling assumptions. A parsimonious annual model then fits

$$\widehat{W}_y = s\, W_y^{evap}$$

or related energy/evaporation regressions to annual Meta withdrawal.

Interpretation: W^{evap} is a physical mechanism, not total campus withdrawal. The fitted scale absorbs unobserved non-cooling use, blowdown/reuse differences, architecture changes, and model error.

5.4 Facility electricity → regional grid, carbon, and generator water

Attributionally,

$$\widehat{CO}_{2,t} = CI_t E_t^{grid}, \qquad \widehat{W}_t^{ind} = EWIF_t E_t^{grid}.$$

Modern regional demand comes from EIA-930. Historical PACW-West hourly demand is reconstructed using FERC Form 714 monthly West totals plus East+West hourly shape, then validated against EIA-930 in overlap years.

Interpretation: these are regional accounting/context layers. They do not identify Meta's exact serving generators or marginal grid response.

5.5 Water-source pumping → groundwater state

The eventual coupled model would use source-resolved groundwater withdrawal

$$q_{n,\tau}^{dc} = \theta_{n,\tau}^{gw} W_\tau^{dir,with},$$

inside a reduced groundwater state model such as

$$h_{\tau+1} = Ah_\tau + B_R R_\tau - B_Q q_\tau + \varepsilon_\tau.$$

Prineville today: source shares and physical aquifer parameters are not identified. What we now have is OWRD pumping plus QC-qualified GWIS groundwater observations, which is enough to test a much simpler empirical response benchmark first.

5.6 Simulation and future decisions

The stochastic workload module generates plausible workload/utilization shapes,

$$\text{synthetic arrivals} \to u_t \to P_t^{IT} \to \text{energy / cooling / water / carbon scenarios.}$$

Simulation is therefore an **experiment engine**: it can explore interactions under explicit assumptions, but it cannot recover missing historical telemetry or establish causal groundwater impacts.

6 How the literature maps to the model

The project is deliberately not trying to implement every "state-of-the-art" model at once. The relevant question is which literature structure is justified by the data at each layer.

7 What we have learned so far

1. The case is now empirically rich enough to ground the research question

Meta Prineville grew from a comparatively small early campus to a facility using electricity at the TWh/year scale. Reported campus electricity covers 2011–2024 (71,000 MWh to 1.73 TWh). Water withdrawal is reported from 2014. Location-based Scope 2 is reported from 2012; 2011 is not separately disclosed. Annual campus electricity, water withdrawal, and location-based Scope 2 evolve differently, which already shows that one single "size" proxy cannot explain every externality.

2. Annual electricity is much better identified than internal hourly operation

Reported Meta electricity provides strong annual scale, but public data do not reveal the true hourly workload, IT load, cooling mode, or facility component powers. The hourly electricity decomposition is therefore useful for physical/scenario reasoning, not historical telemetry recovery.

3. The current annual water model fails the most important predictive test

Improving the local weather layer did not solve annual water prediction. Training uses observations through 2022; 2023–2024 is a two-year chronological holdout ($N = 2$). The current conditional evaporation-scale model has holdout percentage errors of +128.9% (2023) and +100.6% (2024). A frozen training-mean baseline is closer (+8.6% and −40.4%). Energy-only and evaporation-only candidates also fail to beat those simple historical references.

With only two holdout years, this is a small statistical sample. The practical conclusion is nevertheless clear:

the current annual water model is a mechanism / falsification baseline, not a reliable predictor.

The failure is informative: energy plus simple evaporative weather physics is not sufficient to explain campus withdrawal. Missing drivers may include technology/mode, source/reuse strategy, blowdown, non-cooling use, or other operational changes, but the current data do not identify which one dominates.

4. Public regulatory data can reconstruct missing regional grid context

The FERC-constrained historical PACW proxy agrees well with EIA-930 during the mature 2016–2018 overlap ($r = 0.919$, NRMSE $= 0.083$). This makes pre-EIA regional demand usable as a *validated proxy*, not as directly observed West-hourly demand.

5. Independent permit evidence proves that facility technology is not temporally static

Crook County records document a PRN1 addition of approximately 82.7k ft^2 (reported range 82,273–82,736 ft^2), IT-serving chilled-water/CRAH/chiller infrastructure, late-2023 commissioning/testing, and an additional chiller operational in early 2024.

This is important because a single cooling architecture over 2011–2024 is a simplification. It is **not** evidence that the permit event caused the 2023–2024 water-model failure, and it was not used to retune the holdout.

6. Groundwater has crossed from "missing state" to "testable empirical layer"

The GWIS QC step processed 812 records, of which 800 contain numeric BLS measurements. After metadata-based exclusions, 796 are retained as potentially usable groundwater-state observations. Of these, 427 are explicitly STATIC and 369 are retained with ambiguous/unknown status.

Known problematic observations (for example PUMPING, FLOWING, INJECTING, DRY, or NOT MEASURED) are excluded only when metadata justify exclusion; surprising values are not deleted by magnitude alone.

The target is now physically signed:

$$\text{bls\_anomaly}_{i,t} = BLS_{i,t} - \overline{BLS}_i, \qquad \text{head\_anomaly}_{i,t} = -\text{bls\_anomaly}_{i,t}.$$

Three systems remain *estimation candidates* in the current Task-1 screen—meaning sufficient data to attempt a validated empirical response benchmark, not identified groundwater dynamics:

$$\boxed{\text{Heliport (SRC-GC), Millican (SRC-JA), Airport \#2 (SRC-GB)}}.$$

Airport pumping remains the accepted combined SRC-GA+SRC-GB group. The broad parameterized-aquifer model remains Class B / not ready. No groundwater-response model is fitted.

However, Heliport and especially Millican retain large REPORTED/UNKNOWN excursions that are not explained by available local method/status metadata. Measured groundwater observations are now available and several pumping/well systems contain enough overlap to attempt an empirical benchmark, but measurement-condition uncertainty remains material at some wells. The data are therefore *usable with explicit sensitivity analysis*, not "fully cleaned."

8 What should we trust, and what remains unidentified?

Confidence ladder

Missing information by the scientific relationship it blocks

9 What simulation can and cannot do today

Simulation is useful when its role is explicit.

Useful now:

- explore plausible workload shapes and utilization patterns;
- propagate weather through a simplified cooling model;
- compare explicit subannual water-allocation scenarios;
- study timing interactions with regional carbon signals;
- test future counterfactual technology/source/operating assumptions.

Not justified now:

- recovering Meta's historical workload or hourly IT telemetry;
- reconstructing Meta campus-total monthly water without an identified campus-total meter;

- mapping City-metered service water to Meta annual withdrawal, sewer return, and municipal wells;
- interpreting simulated pumping as the cause of an observed groundwater movement;
- treating scenario output as evidence of an actual historical cooling mode or source share.

10 Immediate next scientific decision

This section specifies a planned experiment; it is not a current fitted result. The next step should answer one narrow empirical question:

Does antecedent pumping add stable chronological predictive information about groundwater levels beyond seasonality and trend?

To align monthly OWRD pumping with uneven GWIS measurement density, the first benchmark should use one target per well-month (for example, the median eligible head anomaly in that month).

Two samples should be pre-specified:

Primary: all metadata-eligible observations,

Sensitivity: explicitly STATIC observations only.

For SRC-GC, SRC-JA, and SRC-GB (with Airport pumping kept as the accepted combined SRC-GA+SRC-GB group), compare:

$$M_0 : h_{i,m} = \alpha_i + s_i(month_m) + \gamma_i m + \epsilon_{i,m},$$

against

$$M_1 : h_{i,m} = \alpha_i + s_i(month_m) + \gamma_i m + \beta_i X_{i,m}^{antecedent\,pumping} + \epsilon_{i,m}.$$

Use only completed prior-month pumping exposures; no same-month total that can contain pumping after the groundwater measurement. Validation should be chronological, not random.

Decision rule:

stable incremental out-of-sample skill	$\Rightarrow$ build a small reduced-order groundwater module,
no stable incremental skill	$\Rightarrow$ add recharge / regional/background controls before model complexity.

Only after the response models are credible should the project move toward coupled workload/source/siting optimization.

11 Discussion questions for the meeting

The most useful advisor feedback is not on implementation detail, but on the modeling hierarchy:

1. Is the overall causal chain the right research abstraction: compute $\rightarrow$ facility $\rightarrow$ grid/water $\rightarrow$ environmental state $\rightarrow$ decision?

2. Is Prineville now sufficiently grounded to serve as the empirical validation case, despite missing internal telemetry?
3. Should the next identification effort prioritize groundwater response, or City/Meta water-boundary reconciliation (sewer, wells, source allocation)?
4. Is the proposed progression—simple empirical response → reduced-order state model → later physical/optimization coupling—the right level of complexity?
5. Which eventual decision problem is most scientifically valuable: operations, source selection, siting, or multi-sector water allocation?

A Complete source traceability

The table below is generated from the current pipeline source catalog (50 products). It is reference material, not meeting narration.

Table 8: Complete source-product inventory from the current pipeline catalog. URLs are repository metadata; missing or non-URL acquisition notes are marked accordingly.

Provider / entity	**Dataset / product**	**URL or acquisition route**	**Coverage / resolution**	**Physical or accounting boundary**	**Model role**	**Main limitation**
Meta / Facebook	Facebook Sustainability Data 2014	https://sustainability.atmeta.com/asset/2014-sustainability-data-disclosure/	2011-2014 vintage audit / annual / Prineville campus (site table)	Prineville campus (site table)	historical source-vintage audit; first 2014 water disclosure	Not the canonical later-year series; site totals are not hourly IT telemetry.
Meta / Facebook	2015 Sustainability Data Disclosure	https://sustainability.atmeta.com/asset/2015-sustainability-data-disclosure/	2011-2015 vintage; 2012 location Scope 2 used in canonical table / annual / Prineville ca...	Prineville campus (site table)	source-vintage audit; supplies 2012 location Scope 2	Annual corporate site table, not meters.
Meta / Facebook	2016 Sustainability Data Disclosure	https://sustainability.atmeta.com/asset/2016-sustainability-data-disclosure/	canonical early electricity (2011-2016); early water; Scope 2 / annual / Prineville campu...	Prineville campus (site table)	canonical early electricity, water, location Scope 2, operational GHG	Site annual totals. Do not treat as hourly IT workload.
Meta / Facebook	Facebook Sustainability Data 2019	https://sustainability.atmeta.com/asset/fb_sustainability-data-disclosure-2019/	canonical 2017-2019 electricity; 2015-2019 water; Scope 2 / annual / Prineville campus (s...	Prineville campus (site table)	canonical mid-period electricity, water, location Scope 2	Site annual totals. Operational GHG definitions are vintage-sensitive.
Meta	Meta 2025 Environmental Data Index (FY2024)	https://sustainability.atmeta.com/wp-content/uploads/2025/10/Meta_2025-Environmental-Data-Index.pdf	2020-2024 electricity, water withdrawal, location Scope 2, operational GHG / annual / Pri...	Prineville campus (site table)	canonical recent campus electricity, withdrawal, location Scope 2	Withdrawal is the disclosed site water quantity; consumption vs withdrawal is not separat...

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Provider / entity	Dataset / product	URL or acquisition route	Coverage / resolution	Physical or accounting boundary	Model role	Main limitation
Meta	Meta 2023 Environmental Data Index (methods)	https://sustainability.atmeta.com/wp-content/uploads/2023/07/Meta-2023-Environmental-Data-Index.pdf	WUE / withdrawal / consumption estimation definitions / methods through 2022 / company/si…	company/site methodology	definition/methods context for WUE and water accounting	Canonical numeric series come from 2016/2019/2025 vintages via build_targets.py, not this…
Meta Engineering	Designing a Very Efficient Data Center (2011-04-14)	https://engineering.fb.com/2011/04/14/core-infra/designing-a-very-efficient-data-center/	2011 initial full-load PUE 1.07 and WUE 0.31 L/kWh; outside-air evaporative cooling; no c…	Prineville initial design	epoch-1 physics prior and PUE/WUE falsification point	Design point, not measured annual PUE/WUE. Gray-box fan/other fractions are code priors,…
Meta	Facebook's Prineville Data Center is Growing Again	https://datacenters.atmeta.com/2021/03/facebooks-prineville-data-center-is-growing-again/	2010 groundbreaking; 2021-03-18 expansion announcement (~900,000 sq ft; 11 buildings / ~4…	campus	candidate capacity-state annotation only; completion dates not identified	Announcement is not a commissioning date or MW capacity series.
City of Prineville	Economic Opportunities Analysis (2026 draft)	https://cityofprineville.com/sites/default/files/fileattachments/city_council/page/19317/final_draft_prineville_eoa_january_22_2026.pdf	2010-2024; 11 data-center buildings constructed / planning/event / city / campus	city / campus	end-state campus-scale cross-check	Does not identify individual commissioning dates or MW.
City of Prineville	City Water/Sewer system description	https://www.cityofprineville.com/1426/WaterSewer	current / current description / municipal system	municipal system	context for municipal ownership, wells/reservoirs, metering	Not a production time series. Monthly Meta delivery is not in this pipeline.
City of Prineville	2024 Consumer Confidence Report	https://www.cityofprineville.com/DocumentCenter/View/516/2024-Consumer-Confidence-Report	2024; groundwater-only source statement / annual/system / PWS 00682	PWS 00682	named well inventory / groundwater-only statement	Not monthly production. Not Meta campus delivery.
City of Prineville	Facebook utility-meter package (2026 delivery)	local data/raw/city_prineville_public_records_2026/	2012–2026 meter monthly; 2026 partial through July	src/prepare_city_prineville_utility.py	observed City-metered Facebook Data Center WATER-COMM + ADD'L WATER; not Meta campus-total withdrawal	SWR / WELL METER FOR SEW / bulk / Trailer City / Warehouse remain unresolved identities.
Oregon Health Authority / Drinking Water…	Prineville City PWS 00682 inventory	https://yourwater.oregon.gov/inventory.php?pwsno=00682	official source IDs, well-log IDs, source status / current/historical status / source/fac…	source/facility inventory	municipal well/source identity for OWRD mapping and HUC12 join	13 wells unresolved (no official coordinates). Yancey Well #3 maps outside study HUC geog…

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Provider / entity	Dataset / product	URL or acquisition route	Coverage / resolution	Physical or accounting boundary	Model role	Main limitation
Oregon Water Resources Department	Water Use Report Query entity exports	https://apps.wrd.state.or.us/apps/wr/wateruse_query/	City entity WY2010-2025; Vitesse/Facebook WY2011-2024 bundled / monthly where reported /...	POD / Report-ID	external municipal production and Meta/Facebook-associated direct POD evidence;...	City production is not Meta delivery. Direct POD is not total Meta withdrawal. Missing mo...
Oregon Water Resources Department	Water Use Reporting program rules	https://www.oregon.gov/owrd/programs/WaterRights/Reporting/WUR	current / methods		reporting-requirement context	Not a numeric series.
Oregon Water Resources Department	Aquifer Storage and Recovery program page	https://www.oregon.gov/OWRD/programs/GWWL/GW/Pages/ASR.aspx	current / methods/system		ASR monitoring/accounting context only	No operational ASR injection/recovery series or groundwater-head model is implemented.
Oregon Water Resources Department / City...	Prineville ASR Dedicated Well #1/#2 grant application (2020)	https://www.oregon.gov/owrd/programs/FundingOpportunities/WaterProjectGrantAndLoans/Documents/Applications%20Received%202020%20Cycle/PrinevilleASR_Application.pdf	states 260 MG/y additional storage as application context / 2020 application / historical...	City ASR project	groundwater-system intervention context	Not a calibrated head/storage time series. Catalogued application PDF is still not presen...
Oregon Water Resources Department / City...	Prineville ASR 2020 attachments / 2018 feasibility study	https://www.oregon.gov/owrd/programs/FundingOpportunities/WaterProjectGrantAndLoans/Documents/Applications%20Received%202020%20Cycle/PrinevilleASR_Attachments.pdf	hydrographs, aquifer properties, ASR pilot design document context; numeric heads now...	local aquifer / ASR pilot	ASR engineering context; hydrographs not digitized because GWIS tabular levels...	Catalogued attachments PDF is still not present locally. Local Crook County permit PDFs w...
Oregon Water Resources Department	Groundwater Information System (GWIS) well, construction, lithology,...	OWRD GWIS exports placed under data/raw/gwis_data_new/; no additional download in this pipeline stage	local export of eight unique wells after file-hash deduplication; numeric BLS/AMSL where...	individual wells (official GWIS coordinates where pres...	measured groundwater-level observations and official well identity/construction...	Duplicate raw exports are provenance-only. Mixed NGVD1929/NAVD1988 datums are not convert...
NOAA NCEI	Integrated Surface Database / Global Hourly product family	https://www.ncei.noaa.gov/products/land-based-station/integrated-surface-database	product family; Prineville uses KRDM files below / hourly/subhourly / station	station	product documentation for the weather driver	source_manifest still says station-ID verification needed; executable config uses KRDM 72...
NOAA NCEI	Global Hourly yearly station files: 72692024230 (KRDM / Roberts Field)	https://www.ncei.noaa.gov/data/global-hourly/access/{YEAR}/72692024230.csv	2011-2024 reconstruction window in processed weather_hourly.csv / hourly after processing...	KRDM station (Redmond Roberts Field; not on-campus)	exogenous weather driver for gray-box cooling/evaporation	Off-campus proxy for campus microclimate. source_manifest says files are not bundled; pro...

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Provider / entity	Dataset / product	URL or acquisition route	Coverage / resolution	Physical or accounting boundary	Model role	Main limitation
National Weather Service	Roberts Field / KRDM time series page	https://www.weather.gov/wrh/timeseries?site=KRDM	station identity confirmation / current / station metadata	station metadata	confirm nearby reference station KRDM	Not the numeric archive used by prepare_weather.py.
NOAA MADIS	KS39 / Prineville Airport METAR (MADIS)	MADIS mesonet METAR; local cache data/raw/noaa_madis_ks39/	QC-usable KS39 preferred in canonical weather from 2015-09-01 local; earlier KS39 is audi…	KS39 station (Prineville Airport; not on-campus)	preferred near-site weather driver from 2015-09-01 local when MADIS QC-usable	Not campus microclimate. MADIS QC is not relaxed. 2016 pressure fallback is genuine NOAA…
U.S. Geological Survey	National Water Availability Assessment Data Companion API	https://api.water.usgs.gov/nwaa-data/data	model-specific (see product rows) / monthly / HUC12	HUC12	acquisition route for HUC12 water-use and IWA products	Not Meta meters. No CONUS dump. Do not use after documented product end dates.
U.S. Geological Survey	IWA assessment outputs CONUS 2025	https://water.usgs.gov/nwaa-data/config/api/model/iwa-assessment-outputs-conus-2025.json	2009-10 through 2020-09 / monthly / HUC12 (site 170703051002; local 9; same-HUC8 52)	HUC12 (site 170703051002; local 9; same-HUC8 52)	regional IWA streamflow, cumulative consumption, availability identity, SUI	availab = strflow - consum is an internal identity, not independent hydrologic validation…
U.S. Geological Survey	Public-supply consumptive-use model (pscutot)	https://water.usgs.gov/nwaa-data/config/api/model/wu-public-supply-cu.json	2009-01 through 2020-12 / monthly / HUC12	HUC12	modeled public-supply CU context; not Meta-specific	Do not add pscutot into IWA consum. Missing after 2020-12.
U.S. Geological Survey	Public-supply withdrawals model (pswdtot/pswdgw/pswdsw)	https://water.usgs.gov/nwaa-data/config/api/model/wu-public-supply-wd.json	2000-01 through 2020-12 / monthly / HUC12 / HUC8 plus extra hydro-near HUC12	HUC12 / HUC8 plus extra hydro-near HUC12	modeled public-supply withdrawal context	Not City OWRD production and not Meta withdrawal. Missing after 2020-12.
U.S. Geological Survey	Crop irrigation withdrawals model (irrwdtot)	https://water.usgs.gov/nwaa-data/config/api/model/wu-irrigation-wd.json	2000-01 through 2020-12 / monthly / HUC12	HUC12	modeled irrigation withdrawal context	Competing-sector context only. Missing after 2020-12.
U.S. Geological Survey	Crop irrigation consumptive-use model (irrcutot)	https://water.usgs.gov/nwaa-data/config/api/model/wu-irrigation-cu.json	2000-01 through 2020-12 / monthly / HUC12	HUC12	modeled irrigation consumption context	Do not add irrcutot into IWA consum. Missing after 2020-12.
U.S. Geological Survey	Watershed Boundary Dataset HUC12 (The National Map)	https://hydro.nationalmap.gov/arcgis/rest/services/wbd/MapServer/6/query	study HUC geography / current WBD service vintage used in the download / HUC12 polygon	HUC12 polygon	study HUC geography and municipal well point-in-polygon	Site HUC12 170703051002 is site_point_huc12; campus polygon is unverified. Direct Faceboo…
U.S. Energy Information Administration	EIA-930 / Real-Time Operating Grid	https://www.eia.gov/electricity/gridmonitor/about	bundled workbook 2015-07-01 through reconstruction window 2024-12-31 / hourly / balancing…	balancing authority PACW	regional demand/generation/interchange context; hourly consumed CO2 intensity w…	PACW is not campus electricity. Consumed CO2 intensity begins 2018-07. Not a Meta-specifi…

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Provider / entity	Dataset / product	URL or acquisition route	Coverage / resolution	Physical or accounting boundary	Model role	Main limitation
U.S. Energy Information Administration	PacifiCorp West balancing-authority dashboard	https://www.eia.gov/electricity/gridmonitor/dashboard/electric_overview/balancing_authority/PACW	2015-07-01 onward in bundled workbook / hourly / BA PACW	BA PACW	PACW identity for the EIA-930 extract	Same workbook as EIA930; dashboard is the acquisition route, not a second numeric source.
Federal Energy Regulatory Commission	Form 714 Annual Electric Balancing Authority Area and Planning Area R...	https://www.ferc.gov/industries-data/electric/general-information/electric-industry-forms/form-no-714-annual-electric	2011-2018 annual filings present under data/raw/ferc_form_714/ / monthly West NEL/ops; ho...	PacifiCorp-West BA (monthly); PacifiCorp East+West com...	2011-2018 reported PacifiCorp-West monthly demand/operations; East+West hourly...	East+West hourly is not PACW-West. FERC does not provide consumed CO2 intensity, fuel mix...
U.S. Environmental Protection Agency	eGRID detailed US-customary workbooks	https://www.epa.gov/egrid	eGRID2010-2023 workbooks covering model years 2011-2024 / annual vintage (lagged map for...	eGRID subregion NWPP (ZIP 97754)	methodology/accounting consistency benchmark = Meta MWh × NWPP output ra...	Nonhourly campus intensity, not PACW demand, not market/REC accounting. 2024 uses eGRID2...
U.S. Environmental Protection Agency	Power Profiler ZIP Code Tool v14.2	https://www.epa.gov/system/files/documents/2025-06/power_profiler_zipcode_tool_v14.2.xlsx	ZIP 97754 uniquely NWPP; Subregion 2/3 blank / eGRID2023 geography / ZIP 97754 → NWPP	ZIP 97754 → NWPP	campus consumption-location subregion assignment	Geography lookup only; rates come from eGRID vintages.
Oregon Department of Energy	Biennial Energy Report, Renewable Energy chapter	https://www.oregon.gov/energy/Data-and-Reports/Documents/BER-Chapter-3-Renewable-Energy.pdf	2018 PacifiCorp Schedule 272 unbundled REC agreement / policy/event 2018 / PacifiCorp / F...	PacifiCorp / Facebook contract context	market-vs-location carbon accounting breakpoint annotation	Does not change location-based eGRID/PACW physical accounting.
U.S. EPA CAMPD	Oregon hourly emissions extract (CEMS)	EPA CAMPD custom Oregon hourly extract bundled under data/raw/campd/; UUID in filename hourly-emissions-b0f0572b-0f0b-48a9-99c2-8563b9cdc37...	2011-2024 in processed campd_or_unit_hourly.csv (QC: years 2011-2024 present) / hourly /...	Oregon CAMD Facility ID × Unit ID	Oregon generator emissions/generation; no site-to-generator attribution	Does not identify which generators served Meta Prineville. Posted CO2/SO2/NOx mass are no...
U.S. EPA / EIA	EPA/EIA unit crosswalk	Bundled at data/raw/epa_eia_crosswalk/epa_eia_crosswalk.csv. Exact download URL is not stored in source_manifest.csv.	used to join Oregon CAMD units; unmatched units retained / crosswalk vintage as bundled...	CAMD unit ↔ EIA plant/generator/boiler	join CAMD units to EIA identifiers without exploding emissions	Cardinality includes one_to_many and many_to_many; emissions are not copied onto generato...

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Provider / entity	Dataset / product	URL or acquisition route	Coverage / resolution	Physical or accounting boundary	Model role	Main limitation
U.S. Energy Information Administration	EIA Form 860 annual generator inventory	Annual archives bundled under data/raw/eia860/ as eia860{year}.zip. Form documentation lives at EIA; a specific URL is not stored in source…	2011-2024 model years / annual / Oregon plants/generators after filter	Oregon plants/generators after filter	Oregon generator characteristics / operating-retirement dates	National archives filtered to Oregon after read. No campus attribution.
U.S. Energy Information Administration	EIA Form 923 generation and fuel	Annual archives bundled under data/raw/eia923/. A specific URL is not stored in source_manifest.csv.	2011-2024 / monthly where respondent frequency=M; annual-only respondents flagged / Orego…	Oregon plants after filter	Oregon net generation/fuel; CAMPD vs EIA-923 reconciliation diagnostic	Annual respondents' published monthly Netgen is not a monthly observation. No campus attr…
U.S. Energy Information Administration	EIA standardized cooling-detail / Schedule 8 cooling operations	Bundled under data/raw/eia_cooling/. A specific URL is not stored in source_manifest.csv.	standardized cooling water used for 2014-2024; 2013 Schedule 8 volumes where reported in…	Oregon cooling systems after filter	Oregon thermoelectric cooling water; generator-side water intensity where gener…	2011-2012 native units not converted. Negative official consumption is not clipped and is…
Oregon Department of Environmental Quality	Vitesse LLC (Meta/Facebook Prineville) air permit 07-0037-ST-01	Collected PDFs under data/raw/deq_air/ (permit 07-0037, 2012-2025). A public DEQ catalog URL is not stored in source_manifest.csv.	text-extractable ARs include 2012, 2014-2017, 2020, 2022-2023; scan-only 2019/2021 ARs no…	onsite backup generators at the Prineville campus	onsite backup generation/emissions independent of grid Scope 2	Scan-only PDFs unused. Proposed units are not treated as active. Nameplate MW is not IT c…
Oregon Department of Environmental Quality	Oregon GHG electricity-delivery workbooks	Workbooks under data/raw/deq_ghg/. SOURCE_INSTRUCTIONS.md process ghgElectricityEms.xlsx for Pacific Power (PacifiCorp) Oregon deliveries.	Pacific Power sheet processed; other GHG workbooks treated as provenance unless they cont…	Pacific Power Oregon deliveries (not campus)	utility-territory GHG context; not Meta location Scope 2	Not campus electricity and not onsite backup. Kept separate from eGRID/PACW.
Crook County ePermitting / inspection sum…	Vitesse/Meta Prineville building permit inspection-summary PDFs	Manual collection of Inspection Summary Report PDFs curated into data/canonical/campus_permit_evidence.csv. County portal URL is not stored…	2010 onward inspection summaries in the curated table / event / opened and final-or-CO da…	campus parcels / building_id	capacity-epoch and commissioning-support chronology; PRN1 2023/2024 transition…	Opened date is an ePermitting proxy, not necessarily permit issuance. PRN1 addition area…

B Complete quantity / model traceability

The current registry contains 82 glossary quantities and 26 registered methods/models. Implemented quantities are listed first; quantities that remain NOT IDENTIFIED are grouped compactly; methods are classified by role so that accounting identities, reconstructions, fitted predictors, simulations, and QA screens are not all read as predictive models.

Table 9: Implemented scientific quantities (current quantity registry). Provenance uses the report vocabulary; catalog class `simulated` is mapped to SCENARIO.

Quantity / symbol	Meaning	Unit	Boundary	Provenance	Source or equation	Model block	Status / validation
Workload arrivals ($A_{c,t}$)	New work arriving in interval t.	scenario work counts (not kWh)	Generative Cox-process arrivals; not campus scheduler telemetry.	SCENARIO	none (no telemetry); src/stochastic_conditional_simulation.py	workload/IT	implemented as generative scenario only; low (scenario, not observation)
Executed workload ($x_{i,c,\tau,t}$)	Work executed in t from a stable aggregate queue.	scenario work units	Scenario queue service; not observed executed work.	SCENARIO	src/stochastic_conditional_simulation.py	workload/IT	implemented as generative scenario only; low
Utilization ($u_{i,t}$)	Share of usable capacity that is active.	index (scale-free)	Scale-free utilization index in the stochastic proxy only.	SCENARIO	src/stochastic_conditional_simulation.py	workload/IT	scenario index only; low
Backlog ($J_{c,t}$)	Unfinished work in the stochastic aggregate queue.	scenario work units	Scenario aggregate queue backlog.	SCENARIO	src/stochastic_conditional_simulation.py	workload/IT	implemented as generative scenario only; low
IT power ($P^{IT}_{i,t}$)	Electricity used by IT equipment.	MW	Latent IT power: annual scale fitted to campus facility electricity, or scenario shape th…	FITTED	Meta annual facility electricity + NOAA weather + gray-box	workload/IT	implemented as fitted latent (conditional) / simulated-then-closed (stochastic); low for…
IT energy ($E^{IT}_{i,t}$)	Hourly IT power summed to MWh.	MWh	Integral of latent IT power; not a measured IT-energy series.	DERIVED	fitted P_IT	workload/IT	implemented as derived from fitted P_IT; low-medium
Weather state (T^{db}, RH, p, T^{wb})	Canonical KS39/KRDM hourly weather for local calendar years 2011-2024.	degC; percent; Pa	Canonical KS39/KRDM station weather used as the campus cooling driver; stations are not o…	MEASURED	NOAA_GH_KRDM_FILES / NOAA_MADIS_KS39	weather/cooling	implemented; high as station observations; medium as campus driver
IT heat ($Q^{IT}_{i,t}$)	Operational-scale IT heat.	MW_th (implicit)	Q_IT ≈ P_IT inside the gray-box; latent IT boundary.	DERIVED	fitted P_IT	weather/cooling	implemented as physics identity inside gray-box; medium as an accounting assumption; unva…
Cooling load ($Q^{cool}_{i,t}$)	Heat that must be removed.	MW_th (implicit)	Cooling load approximated by IT heat; campus modeled, not metered.	DERIVED	gray-box	weather/cooling	partial: Q_cool ≈ Q_IT baseline; low-medium
Thermal state vector ($T_{i,t}$)	Rack/inlet/GPU temperatures.	degC	Modeled supply-air temperature only.	DERIVED	gray-box + weather	weather/cooling	partial (t_supply_C only); low

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Quantity / symbol	Meaning	Unit	Boundary	Provenance	Source or equation	Model block	Status / validation
Cooling technology/mode ($m_{i,t}$)	Gray-box discrete mode: outside_air_or_winter_mix / partial_evap / full_evap from weather vs 25 C target.	categorical	Weather-rule evaporative mode at the modeled campus, not BMS logs.	DERIVED	gray-box + canonical KS39/KRDM weather	weather/cooling	implemented as rule-based mode; medium for technology class; low for hourly mode
Cooling control ($u_{i,t}^{cool}$)	Setpoints/fan speeds/flows.	degC / effectiveness	Fixed gray-box setpoints/priors.	SCENARIO	code priors in Params	weather/cooling	partial (fixed priors); low
Cooling power ($P_{i,t}^{cool}$)	Gray-box has no chiller COP.	MW	Fan plus evaporative-aux fractions of latent IT power.	DERIVED	gray-box priors	weather/cooling	implemented as fan + evaporative-aux fractions of IT power; low-medium
Facility power ($P_{i,t}^{fac}$)	P_fac = P_IT + P_fan + P_other + P_evap_aux.	MW	Hourly reconstructed facility power closed to Meta campus annual electricity.	DERIVED	gray-box + annual electricity closure	weather/cooling	implemented; high for annual reported E_fac; low for hourly P_fac
Facility electricity (annual reported) (E^{fac})	Meta-disclosed Prineville annual facility electricity.	MWh	Meta-disclosed Prineville annual facility electricity. Not PACW BA demand.	REPORTED	META_2016_DISCLOSURE / META_2019_DISCLOSURE / META_2025_INDEX via build_targets.py	weather/cooling	implemented; high as disclosed annual total
PUE (PUE)	P_fac/P_IT from the gray-box.	unitless	Derived P_fac/P_IT from the gray-box on the latent IT + reconstructed facility boundary.	DERIVED	gray-box	weather/cooling	implemented as derived KPI; medium vs 2011 design; low as a claim of true hourly PUE
Direct WUE (WUE^{dir})	ISO/IEC 30134-9 WUE = consumption / IT energy.	L/kWh (facility withdrawal intensity);…	Derived withdrawal per facility kWh on the Meta campus table; not ISO WUE.	DERIVED	Meta annual water and electricity	water	partial: withdrawal/facility-kWh derived; ISO WUE unavailable; high for the derived withd…
Raw evaporative water (W^{evap})	Gray-box humidification water from humidity-ratio increase at constant moist-air enthalpy, converted m^3/…	m^3/h (hourly); m^3 (annual sum)	Gray-box raw evaporative humidification at the modeled campus; not makeup or Meta withdra…	DERIVED	gray-box + weather + fitted P_IT	water	implemented; low as a withdrawal estimate; medium as a weather-shaped evaporative index
Direct water withdrawal (Meta annual) ($W^{dir,with}$)	Meta-disclosed Prineville annual water withdrawal.	m^3	Meta-disclosed annual campus withdrawal. Not City production and not Vitesse/Facebook POD…	REPORTED	META_2016_DISCLOSURE / META_2019_DISCLOSURE / META_2025_INDEX	water	implemented; high as disclosed annual withdrawal

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Quantity / symbol	Meaning	Unit	Boundary	Provenance	Source or equation	Model block	Status / validation
Hourly/annual water-withdrawal proxy (con… ($s \times W^{evap,raw}$)	Train-only global multiplicative scale on raw evaporation.	m^3	Predicted/proxy campus withdrawal from gray-box evaporation times a train-only scale; sti…	PROXY	raw evaporation $\times$ fitted scale	water	implemented; low (holdout fails)
Municipal / City production ($CityPODproduction$)	OWRD accepted City of Prineville municipal source-group production.	m^3	OWRD accepted City of Prineville municipal POD production. Not Meta campus delivery.	REPORTED	OWRD_WUR_QUERY	water	implemented; high as OWRD reported City production; not usable as Meta water
Vitesse/Facebook direct POD use (POD64500/64845/64846)	OWRD reported use at Meta/Facebook-associated direct groundwater PODs.	m^3	OWRD Vitesse/Facebook direct groundwater POD reports. Not total Meta withdrawal and not U…	REPORTED	OWRD_WUR_QUERY	water	implemented; high as OWRD POD reports; low as a campus total
USGS irrigation water ($irrwdtot, irrcutot$)	Modeled HUC12 crop irrigation withdrawal and consumptive use.	m^3/month (converted from mgd)	USGS modeled HUC12 irrigation WD/CU. Not campus water.	PROXY	USGS_NWAA_IRR_WD / USGS_NWAA_IRR_CU	water	implemented as regional context; medium as USGS modeled regional context
IWA cumulative streamflow ($strflow$)	USGS IWA cumulative streamflow (upstream+local) in mm and m^3 per month.	mm/month and m^3/month	USGS IWA cumulative streamflow (upstream+local routing). Not local-use WD.	PROXY	USGS_NWAA_IWA	water	implemented; medium as USGS modeled
IWA cumulative consumption ($consum$)	USGS IWA cumulative consumption (upstream+local, not local-only).	mm/month and m^3/month	USGS IWA cumulative consumption (upstream+local). Not pscutot/irrcutot.	PROXY	USGS_NWAA_IWA	water	implemented; medium as USGS modeled
IWA surface-water availability ($availab$)	USGS IWA availability field.	mm/month and m^3/month	USGS IWA availability identity strflow-consum. Not independent hydrology.	DERIVED	USGS_NWAA_IWA	water	implemented; high that the identity holds in the files; that is not hydrologic skill
IWA supply-use index (sui)	USGS IWA SUI field for the site HUC12.	fraction (native)	USGS IWA supply-use index on the routed product.	PROXY	USGS_NWAA_IWA	water	implemented; medium as USGS modeled
USGS public-supply withdrawal/CU ($pswdtot, pswdgw, pswdsw, pscutot$)	Modeled HUC12 public-supply water use.	m^3/month	USGS modeled HUC12 public-supply WD/CU. Not City OWRD production and not Meta withdrawal.	PROXY	USGS_NWAA_PS_WD / USGS_NWAA_PS_CU	water	implemented as regional context; medium as USGS modeled

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Quantity / symbol	Meaning	Unit	Boundary	Provenance	Source or equation	Model block	Status / validation
PACW grid demand / interchange / generati... $(EIA-930PACW)$	Balancing-authority operations from the bundled PACW workbook.	MWh	EIA-930 PacifiCorp West balancing-authority operations. Not campus electricity.	REPORTED	EIA930	grid/carbon	implemented; high as BA published data; not campus
FERC PacifiCorp-West monthly NEL and oper... $(NEL_W, G_W, I_W, peak_W, min_W)$	Reported FERC Form 714 PacifiCorp-West monthly net energy for load, net generation, net interchange, monthly...	MWh; MW	Reported FERC Form 714 PacifiCorp-West monthly NEL/ops. Not campus electricity and not ho...	REPORTED	FERC_FORM_714	grid/carbon	implemented; high as FERC-reported BA monthly totals; not campus
FERC PacifiCorp East+West combined hourly... (D_EW, t)	Reported FERC Form 714 Schedule 2 hourly demand for the PacifiCorp East+West combined planning area.	MW	Reported FERC East+West combined planning-area hourly demand. Not PACW-West.	REPORTED	FERC_FORM_714	grid/carbon	implemented; high as reported combined planning-area demand; not West BA hourly
FERC-constrained PACW-West hourly demand... $(Dhat_W, t)$	FERC-only affine reconstruction: within each month, East+West hourly shape scaled to West monthly NEL with no...	MW	FERC-constrained hourly PACW-West proxy. Not reported PACW hourly demand and not campus e...	PROXY	FERC_FORM_714	grid/carbon	implemented as diagnostic/candidate pre-EIA proxy; depends on overlap metrics; keep as di...
Oregon generator generation and emissions (G_g, E_g)	CAMPD gross generation and posted CO_2/NO_x/SO_2; EIA-923 net generation/fuel.	MWh; tonnes	Oregon CAMPD/EIA-923 generation and emissions. Not attributed to Meta and not PACW demand.	MEASURED	EPA_CAMPD_OR / EIA_923	grid/carbon	implemented as regional generator tables; high as Oregon CEMS/EIA tables; none for Meta a...
Generator cooling water (WI_g)	EIA cooling-product water for Oregon plants where reported.	m^3 (from million gallons)	Oregon EIA cooling-product water. Not Meta indirect water.	REPORTED	EIA_COOLING	grid/carbon	implemented as Oregon tables; medium where reported; none as Meta indirect water
Average electricity water intensity factor $(EWIF)$	Generation-weighted cooling-water intensity over Oregon EIA-923 plant-months that are cooling-intensity-eligi...	m^3/MWh	Partial-coverage Oregon cooling-plant EWIF. Missing cooling water is not zero. Not Meta g...	DERIVED	EIA_COOLING	grid/carbon	implemented as partial-coverage regional diagnostic; medium as a cooling-plant partial EW...
Indirect electricity water (W^{ind})	Regional-average proxy = partial-coverage Oregon cooling EWIF × Meta campus MWh.	m^3	Regional-average EWIF × Meta MWh. Not generator-attributed campus water.	PROXY	EIA_COOLING × Meta electricity	grid/carbon	implemented as regional-average proxy only; low as campus indirect water; medium as a lab...

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Quantity / symbol	Meaning	Unit	Boundary	Provenance	Source or equation	Model block	Status / validation
Meta reported location-based Scope 2 ($S2^{loc,reported}$)	Disclosed Prineville location-based Scope 2 (tCO2e) where present.	tCO2e	Meta-disclosed location-based Scope 2 for the Prineville campus inventory.	REPORTED	Meta disclosures	grid/carbon	implemented; high as disclosed inventory; method ≠ eGRID identity
eGRID NWPP physical carbon benchmark ($E^{fac} \times EF^{eGRID,NWPP}$)	Meta annual MWh times eGRID subregion total output emission rate for ZIP 97754 → NWPP.	tCO2 / tCO2e	Meta campus MWh times eGRID NWPP output rate. Same Meta MWh as Q_E_FAC; NWPP geography, n...	DERIVED	EPA_EGRID × Meta electricity	grid/carbon	implemented; high as a reproducible benchmark; medium as equality to Meta's inventory met...
PACW hourly consumed CO2 intensity ($CI^{PACW,consumed}$)	EIA-930 consumed CO2 intensity for PACW where published (from 2018-07).	intensity units as prepared in pacw_hou.	PACW consumed-CO2 intensity. Regional BA shape, not campus telemetry.	REPORTED	EIA930	grid/carbon	implemented as regional shape; medium as BA intensity; none as campus marginal emissions
PACW fuel/import carbon score ($sensitivityproxy$)	Named sensitivity score from fuel mix and interchange when EIA consumed intensity is missing.	index	PACW fuel/import sensitivity score on the BA boundary.	PROXY	EIA930 fuel mix / interchange	grid/carbon	implemented as sensitivity only; low
Onsite backup generation hours ($backuphours$)	DEQ annual-report monthly operating hours for permitted engines where extractable.	hours	DEQ 07-0037 onsite engine hours. Not grid Scope 2 and not IT capacity.	REPORTED	ODEQ_AIR_07_0037	grid/carbon	implemented; medium where text-extractable; missing scan-only years
Onsite backup emissions ($onsitet$)	DEQ-calculated annual-report emissions and separate source-test measurements.	tons as reported in DEQ tables (see pro...	DEQ onsite backup emissions. Kept separate from location Scope 2 / eGRID / PACW.	REPORTED	ODEQ_AIR_07_0037	grid/carbon	implemented; medium where extractable
Operational Scope 1+2 (market-sensitive) ($operationalGHG$)	Meta-disclosed operational Scope 1+2 at the site table.	tCO2e	Meta-disclosed operational Scope 1+2; market-instrument sensitive, still a campus invento...	REPORTED	Meta disclosures	grid/carbon	curated in annual table; not used as physical carbon model; high as disclosed; low as phy...
Spatial crosswalk site → grid → generator → wat... (M)	Partial: ZIP → eGRID NWPP and site point → HUC12 exist.		Partial maps from the campus point/ZIP: ZIP 97754 → NWPP and site point → HUC12. Not a genera...	DERIVED	EPA_EGRID_POWER_PROFILER; USGS_WBD	groundwater basin	partial; high for ZIP → NWPP; medium for point HUC12; none for generator attribution or aqu...

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Quantity / symbol	Meaning	Unit	Boundary	Provenance	Source or equation	Model block	Status / validation
Groundwater observations (h^{obs})	Time-indexed OWRD GWIS measured water levels (ft below land surface) at matched and unmatched wells in the lo…	ft below land surface; ft AMSL where GW…	OWRD GWIS measured water levels at individual wells. Not a network head field and not OWR…	MEASURED	OWRD_GWIS	groundwater	implemented as measured GWIS well-level series; measured well observations; identity inco…
Structural uncertainty terms (ξ)	Stochastic module samples facility-overhead and water-shape mixture priors; this is scenario uncertainty, not…	varies	Stochastic overhead/water-shape priors; scenario uncertainty, not identified ξ.	SCENARIO	src/stochastic_conditional_simulation.py	groundwater	partial (scenario ensemble, not ξ identification); low as a statistical uncertainty model
Generator heat rate (HR)	CAMPD heat input is stored; a campus-coupled heat-rate model is not implemented.	mmBtu (CAMPD heat input) where present	CAMPD posted heat input at Oregon CEMS units. Not a campus heat-rate model.	MEASURED	EPA_CAMPD_OR	grid/carbon	partial (raw heat input present; not a campus HR model); medium as CEMS heat input; none…

Table 10: Quantities that remain NOT IDENTIFIED in the current public-data pipeline. These are kept for glossary completeness, not as filled proxies.

Quantity	Meaning	Why unidentified
Compute capacity	Usable compute capacity.	No hardware inventory, nameplate IT MW time series, or scheduler availability.
Delay and SLA	Job delay and class delay limits.	No job timestamps or SLA constraints in the pipeline.
Ambient heat rejection / reuse	Heat reuse is not modeled.	No recovered-heat meters; reuse not assumed.
a2 fixed-approach / fixed-cold-water WUE curves	Gupta et al.	Code uses air-side evaporative humidification physics instead of a2 tower curves.
Makeup / blowdown / drift	Cooling-tower mass-balance streams.	No cycles of concentration, blowdown, or drift data.
Cycles of concentration	Not identified.	No water-chemistry or blowdown data.
Direct water consumption	Water not returned to the local system.	Site consumption is not publicly reported as a separate campus series in the canonical table.
Consumptive fraction	W_cons / W_with.	Needs consumption and withdrawal on the same boundary.
Source-share vector	Share of withdrawal by municipal / surface / groundwater / reclaimed.	No campus source-share meter. Direct POD is not total withdrawal.
Onsite water reuse	Not identified.	No reuse meters.
Average-day and maximum-day demand	Paper-c planning metrics.	No daily campus withdrawal.
Renewable capacity factor	Site or BA renewable CF for scheduling.	No campus renewable production model.
Marginal water factor	$\partial W_{\mathrm{power}}/\partial E_{\mathrm{load}}$.	No dispatch/commitment model.
Total operational water footprint	Direct consumption + indirect.	Cannot form identified W_oper: campus consumption is not identified, and indirect water is only a regional-average proxy rather than genera…
Marginal emission factor	Not implemented.	Average location factors and a fuel/import score are not MEFs.

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Quantity	Meaning	Why unidentified
Electricity cost	Not identified.	No tariff or bill series.
Water cost	Not identified.	No water-rate series.
Investment / capacity cost	Not identified.	No cost or nameplate-capacity time series.
Water footprint (scarcity-weighted)	Not implemented.	AWARE/Aqueduct factors are not in the pipeline.
Scarcity characterization factor	Not implemented.	IWA SUI is not a scarcity CF.
Water-scarcity footprint	Not implemented.	Do not substitute IWA SUI for WSF or head.
Data-center groundwater extraction	Groundwater-supplied share of campus withdrawal.	Needs source-share and/or well meters on the campus boundary.
Other-sector groundwater pumping	Not implemented as a pumping series.	HUC12 USGS is not a well network.
Groundwater recharge	Not identified.	No recharge series or model.
Groundwater head	Not identified.	Well-level GWIS BLS/AMSL observations exist as Q_GW_OBS. A modeled or spatially continuous hydraulic-head field is not identified. Mixed NG...
Groundwater storage	Not identified.	No aquifer storage model.
Inter-node conductance	Groundwater network not identified.	No reduced-order or MODFLOW coupling.
Siting / location choice	Not implemented.	PSCC siting model not in this repository pipeline.
Equity / burden metrics	Not implemented.	Burden object not defined in code.

Table 11: Registered methods and models. Not all entries are predictive models.

Method / model	Advisor class	What it does	Is it a prediction?
Prineville gray-box air-side physics	accounting / physical identity	Maps a supplied IT-power trace and hourly weather to facility power, PUE, cooling mode, and raw evaporative water.	no
Annual facility-electricity closure (latent IT scale)	reconstruction	For each year, choose one constant IT-power scale so the gray-box annual facility energy equals Meta-reported MWh.	no — exact annual closure by construction
Conditional water: global log-scale on raw evaporation	fitted/predictive model	Fit one multiplicative scale s on training years with observed withdrawal via geometric mean of W_obs/W_raw.	yes for holdout annual water; in-sample for train
Conditional water: one-break log-scale (candidate, not selected)	fitted/predictive model	Optional two-scale model with a break year; 3 effective parameters (two scales + break).	n/a (not selected)
Annual water candidate: energy-only (selected)	fitted/predictive model	No-intercept nonnegative regression of annual withdrawal on reported facility MWh.	yes
Annual water candidate: evaporation-only	fitted/predictive model	No-intercept nonnegative regression of annual withdrawal on ensemble-median raw evaporation.	candidate only
Annual water candidate: nonnegative energy + evaporation	fitted/predictive model	NNLS on E_fac and raw evaporation.	candidate only
Stochastic conditional proxy (mixed Cox)	simulation/scenario	Generative scenario: Cox-process arrivals, AR log-intensity, Poisson counts, Gamma work sizes, aggregate queue, utilization → IT shape, annual facility-energy sc...	only the selected annual water model is a prediction; the rest is scenario/closure

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Method / model	Advisor class	What it does	Is it a prediction?
eGRID NWPP × Meta MWh physical benchmark	validation/QA	Methodology/accounting consistency benchmark: Meta campus MWh times eGRID NWPP output rate versus Meta reported location Scope 2.	no
PACW EIA consumed-CO2 relative hourly shape	reconstruction	Optional relative within-year carbon shape from PACW consumed intensity.	no
FERC-constrained PACW-West hourly proxy	reconstruction	Within each month, uses East+West hourly demand only for shape and West monthly NEL/peak/minimum for level.	no
PACW fuel/import carbon score	validation/QA	Named sensitivity proxy only.	no
Annual piecewise-linear SSE break ranking	identifiability screening	Ranks candidate break years by relative SSE reduction of two linear trends vs one, for electricity, water, and water intensity.	no
IWA availability accounting identity	accounting / physical identity	Checks availab = strflow - consum in USGS IWA files.	no
OWRD external water-evidence comparison	validation/QA	Aligns modeled monthly campus water proxy with City production and direct POD without using them as calibration targets or summing boundaries.	no
Oregon CAMPD/EIA generator QC	validation/QA	Join/coverage diagnostics between CAMPD and EIA-923; cooling-water coverage; unit conversions.	no
Groundwater observation scaffold	accounting / physical identity	Compiles well identities, accepted OWRD pumping groups, GWIS measured water levels, construction/aquifer fields, and hydrogeologic parameter citations.	no
Partial-coverage Oregon cooling EWIF	accounting / physical identity	Generation-weighted withdrawal/consumption intensity over cooling-intensity-eligible plant-months with non-missing water.	no

C Selected literature references

References

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Authority / entity	What it is	What we use	Boundary / why it matters
Meta	Data-center operator	Annual Prineville electricity, withdrawal, location-based Scope 2, selected site/design disclosures	Campus-level ground truth; no public hourly IT or customer-meter telemetry.
NOAA / NCEI / MADIS	U.S. atmospheric observation systems	KS39 and KRDM temperature, dew point, pressure; derived wet-bulb	Local external driver for cooling; station coverage differs by year.
OWRD	Oregon Water Resources Department	Water-use reports, water rights/PODs, well records, GWIS groundwater observations	Official Oregon water authority; municipal production, direct POD pumping, and campus withdrawal are distinct boundaries.
OHA Drinking Water Services	State drinking-water regulator/inventory	Prineville public-water-system source identities	Supports authoritative source/well crosswalks for the municipal system.
USGS	Federal water-science agency	HUC12 water use and Integrated Water Availability context	Regional modeled hydrologic/use context; not Meta-specific and not an aquifer node.
PacifiCorp / PACW	Regional utility / balancing-area context	PACW system identity and demand context	Regional power-system boundary; not campus electricity.
FERC	Federal electricity regulator	Form 714 PacifiCorp-West monthly demand/operations and East+West hourly shape	Historical regulatory evidence used to bridge pre-EIA hourly coverage.
EIA	Federal energy-statistics agency	EIA-930 PACW hourly data; EIA-860/923 generator/fuel/cooling data	Modern grid and generator context.
EPA	Federal environmental agency	eGRID regional emissions factors; CAMPD plant emissions	Regional accounting and plant emissions; not Meta-specific marginal impacts.
Oregon DEQ	State environmental regulator	Backup-generator permits, hours, fuel/emissions evidence	Onsite backup-generation context; permitted capacity is not IT load.
Crook County	Local permitting authority	Structural, electrical, mechanical, plumbing permits and inspections	Documents physical expansion and technology chronology; does not directly reveal IT MW or water use.

Table 2: Major authorities and the physical/accounting boundary each contributes. Appendix A lists every individual source product, URL, coverage, and role.

Step	What happens	Prineville example / rule
1. Gather	Prefer operator, regulator, or official measurement systems.	Meta disclosures, NOAA, OWRD, USGS, EIA/FERC/EPA, DEQ, Crook County.
2. QC	Check dates, units, duplicate records, missing-vs-zero, reporting conventions, and source-specific flags.	GWIS pumping/flowing observations are excluded from the state-model sample only when metadata support exclusion.
3. Crosswalk	Determine whether records refer to the same physical entity and boundary.	Airport wells share combined pumping; unresolved Vitesse PODs are not automatically mapped to nearby GWIS wells.
4. Canonicalize	Keep one defensible representation with provenance.	BLS and AMSL are paired representations of the same GWIS measurement, not two independent observations.
5. Model	Infer only quantities that are not directly observed.	Hourly IT load and Meta campus-total monthly water still require reconstruction. City-metered service water is now observed.
6. Validate	Compare against independent data where possible; separate prediction from accounting closure.	FERC historical demand is checked against EIA-930; exact annual electricity closure is not called a forecast.
7. Scenario / decision	Use uncertain quantities only after their status is explicit.	Synthetic workload scenarios can explore interactions but cannot recover historical Meta telemetry.

Table 3: The pipeline logic.

Layer	Quantity	Source / authority	Status	Model role / current conclusion
Workload	x_t, u_t	none public for Meta	SCENARIO	Synthetic workload used only to create plausible operational shapes.
IT	P_t^{IT}	constrained by Meta annual facility MWh	FITTED/ SCENARIO	Latent internal state; not directly validated as telemetry.
Facility	E_y^{fac}	Meta	REPORTED	High-confidence annual target.
Weather	T, T_d, T_{wb}, p	NOAA	MEASURED/ DERIVED	Strong cooling input layer.
Cooling	P_t^{cool}, Q_t^{cool}	gray-box	DERIVED	Mechanism/scenario quantity.
Campus water	W_y^{Meta}	Meta	REPORTED	High-confidence annual withdrawal target from 2014+.
Evaporation	W_t^{evap}	gray-box	DERIVED	Physical cooling mechanism, not total campus withdrawal.
Municipal/direct pumping	Q_t	OWRD	REPORTED	Water-system / groundwater forcing at explicit reporting boundaries.
Regional water context	HUC12 use / availability	USGS	PROXY	Regional context only; not a campus or aquifer measurement.
Grid demand	D_t^{PACW}	EIA + FERC	MEASURED/ PROXY	EIA later; FERC historical bridge earlier.
Carbon	CI_t	EIA / EPA eGRID	DERIVED/ PROXY	Regional average/accounting context.
Plant impacts	$G_{g,t}, CO_{2,g,t}, W_{g,t}$	EIA / EPA	REPORTED/ DERIVED	Generator-level context; no Meta-specific serving attribution.
Groundwater	$h_{i,t}$	OWRD GWIS	MEASURED	QC-qualified state observations; response relationship not yet identified.
Facility chronology	technology/events	Crook County	REPORTED	Documentary evidence of expansion/cooling regime changes.
Groundwater response	(A, B_Q, β)	not yet estimated	NOT IDENTIFIED	Immediate next scientific benchmark.
Decision variables	workload/siting/source choice	future	SCENARIO	Long-run optimization layer after response models are validated.

Table 4: Master source–quantity–model map. Appendix B provides the complete machine-traceable version.

Literature family	What we inherit	Current Prineville approximation	What would justify the next upgrade?
National/accounting: LBNL; Siddik et al.	Explicit system boundaries, aggregate envelopes, spatial environmental accounting	Site-specific reported annual quantities + regional context	Use mainly as external order-of-magnitude / boundary checks.
Facility PUE/WUE: Lei & Masanet	Climate- and technology-dependent coupled energy/water physics	Simplified weather-driven gray-box	Better evidence on cooling technology, controls, and operating telemetry.
Water footprint: Li et al.; Gupta et al.; Han et al.	Direct vs indirect water; temporal water signals; peak public-water capacity	Annual direct water + partial regional generator-water diagnostics	Monthly campus water, source shares, withdrawal/consumption/discharge, peak delivery.
Operational scheduling: WACE; Google; WaterWise; TAPAS	Workload conservation, spatial/temporal flexibility, carbon/water/thermal constraints	Stochastic scenarios only; no actual workload optimization yet	Real workload flexibility/capacity, validated environmental response signals, prices/costs.
Siting/planning: PSCC project	Cost–carbon–water–equity planning backbone	Long-run target, not current implementation	Credible source-specific water and groundwater response functions.
Groundwater: MODFLOW / hydro-economic models	Pumping/recharge $\rightarrow$ head/storage dynamics; multi-sector allocation	First empirical well-level benchmark only	Recharge/background forcing and sufficient hydrogeologic/response information for reduced-order or physical calibration.

Table 5: Literature-to-model map: what is inherited now, what is simplified, and what evidence would justify more complexity.

Level	Components	Interpretation
High-confidence evidence	Meta annual ground truth; canonical NOAA weather; EIA modern grid; OWRD reported pumping; permit chronology; GWIS observations as observations	Use directly at their stated physical/accounting boundaries.
Validated proxy / reconstruction	FERC historical PACW bridge; annual-closed facility-electricity profile	Useful because constraints/overlap validation are explicit; not raw telemetry.
Mechanism / scenario	cooling gray-box; stochastic workload; monthly water allocations; temporal carbon/water scenarios	Useful for sensitivity and counterfactuals, not historical identification.
Weak predictor	annual campus-water models	No demonstrated holdout advantage over naive references.
Not yet identified	hourly Meta IT; monthly campus meters; exact source shares; pumping-to-head causal response; aquifer dynamics; exact serving generators/marginal impacts; costs/optimization	Requires new identification or data before strong claims/optimization.

Table 6: Current confidence hierarchy.

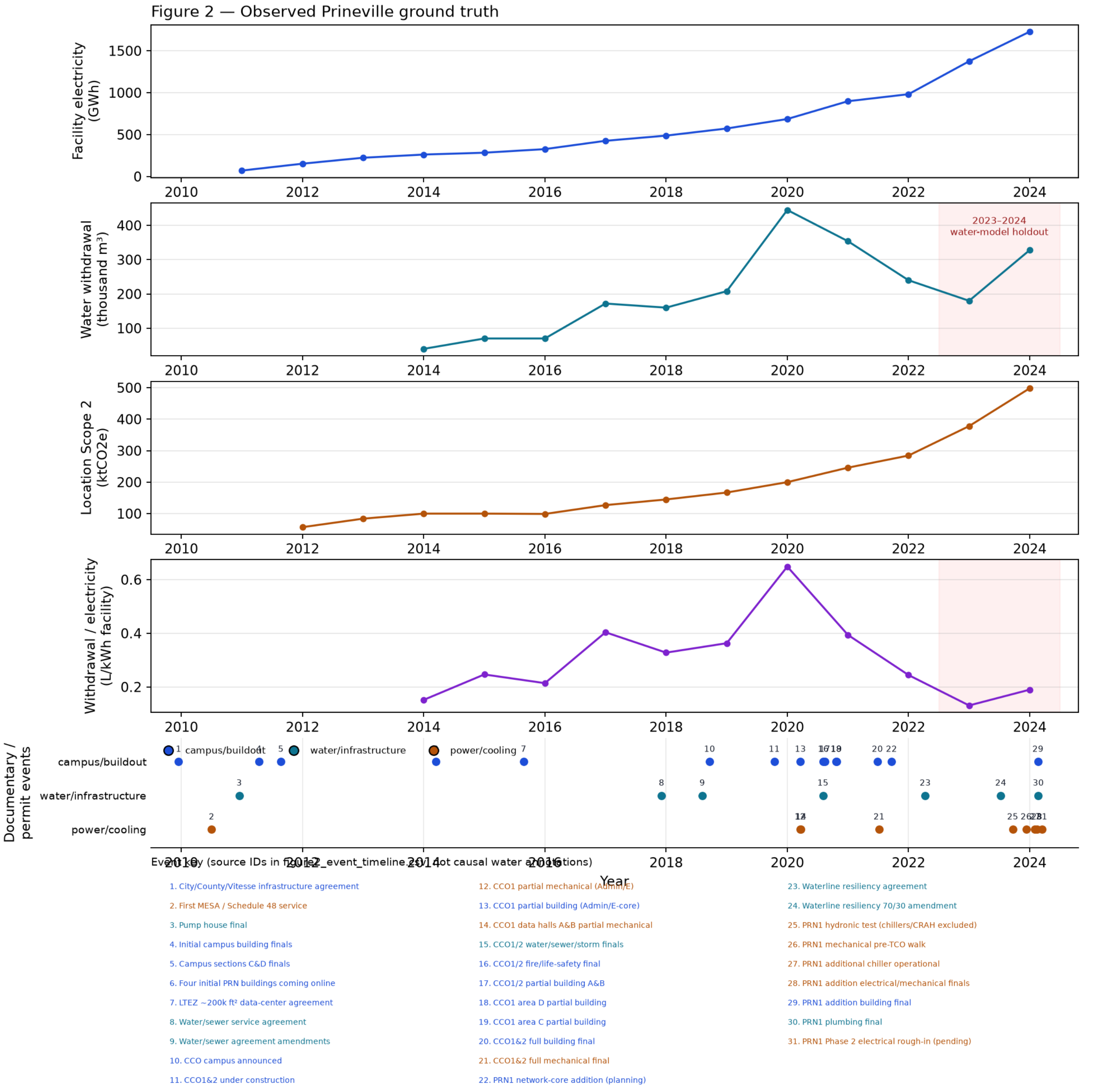
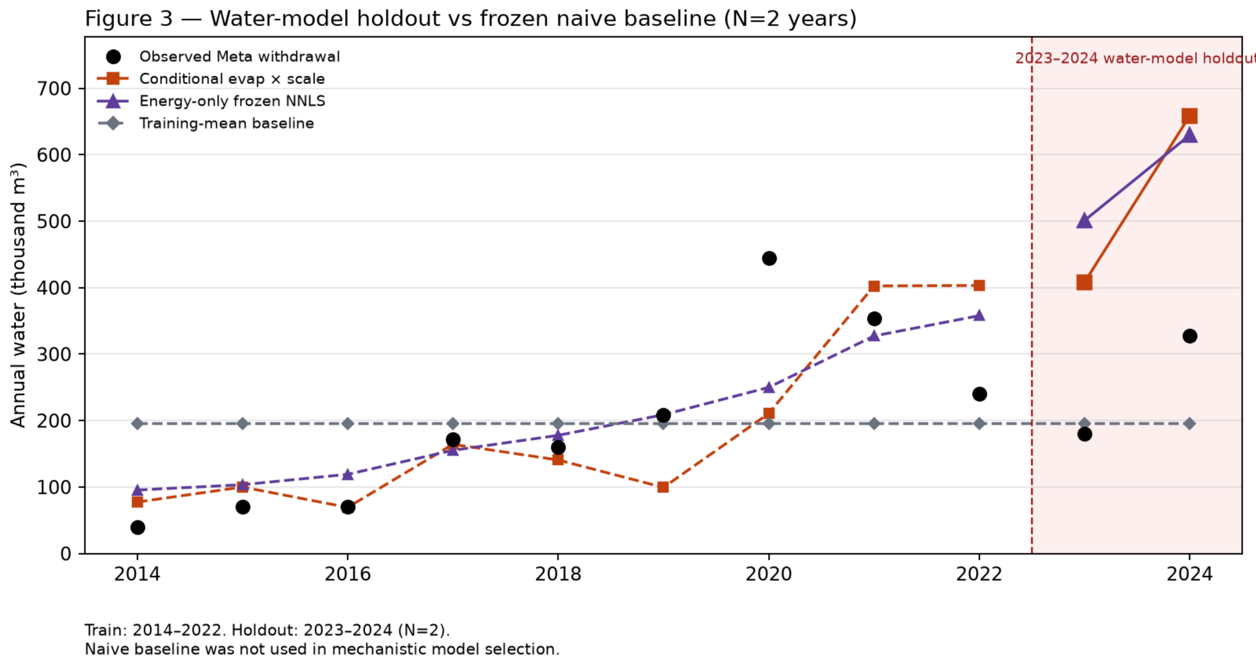


Figure 2: Observed annual Meta Prineville ground truth and intensity evolution. This is the empirical anchor for the case study.



Predictor	Holdout MAPE	2023 % error	2024 % error
Training mean	24.5%	+8.6%	-40.4%
Conditional evap × scale	113.8%	+126.7%	+100.8%
Energy-only NNLS	135.2%	+178.4%	+92.0%

Figure 3: Annual Meta water validation. The main message is comparative out-of-sample skill: current physics/energy predictors do not beat simple historical references on the two held-out years.

Relationship we want	**Main missing / uncertain information**	**What it prevents today**
Workload → IT / facility operation	workload telemetry, IT power, detailed cooling state/control	Historical hourly internal-state identification.
Cooling → campus water	campus-total monthly withdrawal, consumption vs withdrawal, identified discharge/return, source/reuse shares	City-service monthly shape is now observed; total campus/source accounting remains unresolved.
Pumping → groundwater	measurement-condition uncertainty, recharge/background hydrology, other pumping; physical T, S, S_y for full model	Causal/physical groundwater response and aquifer simulation.
Facility load → grid externality	actual supply contract/serving attribution, marginal dispatch response	Meta-specific marginal carbon/water consequences.
Response models → decisions	workload flexibility, tariffs/costs, capacity constraints, validated water/GW response functions	Credible operational/siting optimization.

Table 7: The important gaps are causal/identification gaps, not simply a list of missing files.

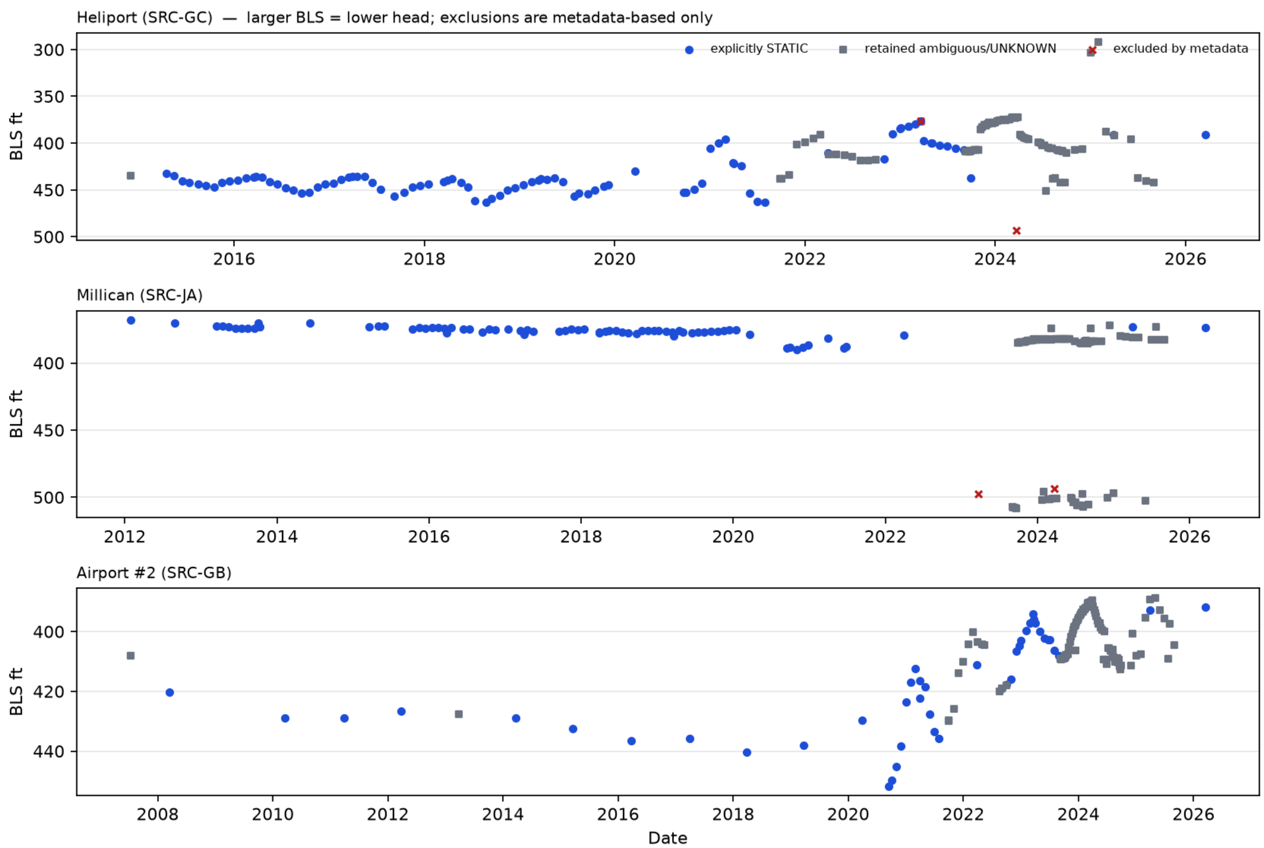


Figure 4: GWIS BLS at the three estimation-candidate wells. Blue circles are explicitly STATIC; grey squares are retained ambiguous/UNKNOWN; red crosses are metadata exclusions (not magnitude outliers). Larger BLS is lower hydraulic head.